SOC CHESS: Intelligent Real-Time Security Operations & AI Threat Detection Platform

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Introduction

- **Modern SOC Bottlenecks:** Enterprise analysts face alert fatigue with 70%+ false positives, siloed network PCAP streams, and disconnected SIEM log indexes.
- **Research Objective:** Architect a local-first cybersecurity platform combining real-time packet capture, enterprise SIEM queries, and dual-tier machine learning.
- **Mathematical Risk Formulation:** Multi-factor risk engine synthesizes ML attack probability, failed attempt counts, and target asset value:

$$Risk\ Score = \min(100.0,\ P(\text{Attack}) \times 80 + \sum w_i \text{Severity}_i + \text{AssetWeight})$$

- Dynamic thresholds: **Low** (0–39), **Medium** (40–69), **High** (70–89), and **Critical** (90–100).

Methodology & Architecture

- **Telemetry Ingestion:** Continuous packet capture via TShark CLI / Wireshark (`socchess://`) plus Splunk Enterprise REST API (Port 8089).
- **Big Data Streaming:** Apache Spark 3.5.3 normalizes 12 raw log formats across 3,708 dataset files (CIC-IDS, UNSW-NB15, KDD-99).

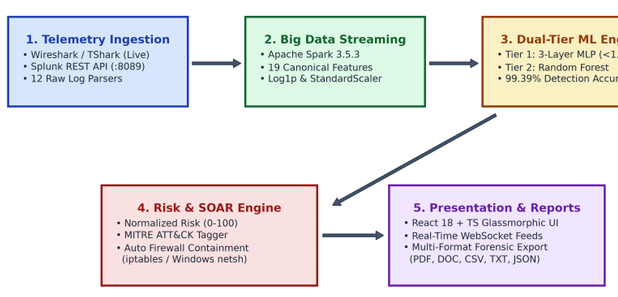


Figure 1. High-level system architecture of the SOC CHESS platform: Ingestion, Spark feature extraction, dual-tier ML, and SOAR containment.

- **19 Canonical Features:** Extracted flow rates, packet lengths, TCP flags (SYN, ACK, PSH, FIN, RST), and window sizes.
- **Pre-processing:** Symmetrical Log1p smoothing $x' = ln(1 + |x|)$ and Z-score standardization.

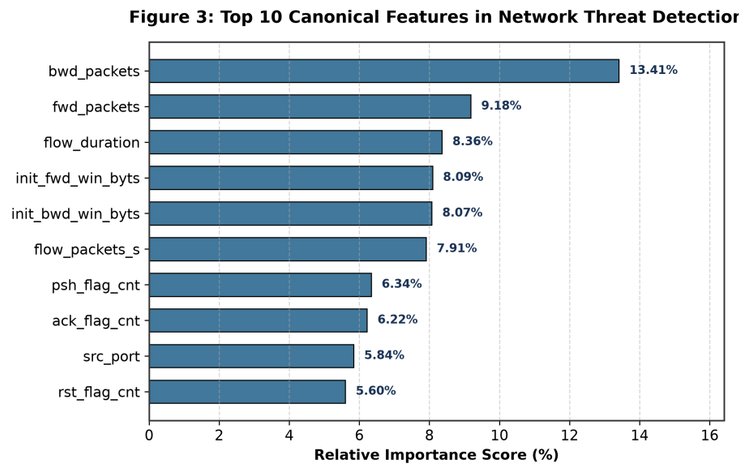


Figure 2. Top 10 canonical flow features ranked by information gain in detecting malicious network traffic.

Dual-Tier AI Detection Engine

- **Tier 1 (Real-Time Stream):** 3-Layer Multi-Layer Perceptron (128 → 64 → 32 ReLU + Softmax) operating at **< 1.2 ms** latency per event.
- **Tier 2 (Batch Champion):** Scikit-Learn Random Forest (200 trees) + XGBoost achieving **99.39% accuracy**.

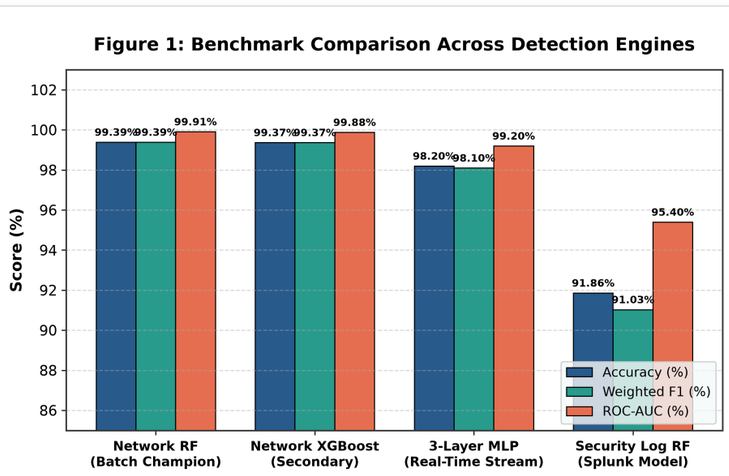


Figure 3. Comparative detection accuracy, weighted F1-score, and ROC-AUC across all four operational model pipelines.

Pipeline	Algorithm	Acc (%)	F1 (%)	ROC-AUC	Latency
Network Batch	Random Forest	99.39%	99.39%	0.9991	0.42 ms
Network Secondary	XGBoost	99.37%	99.37%	0.9988	0.48 ms
Real-Time Stream	3-Layer MLP	98.20%	98.10%	0.9920	0.85 ms
Security Logs	Log RF Classifier	91.86%	91.03%	0.9540	1.10 ms

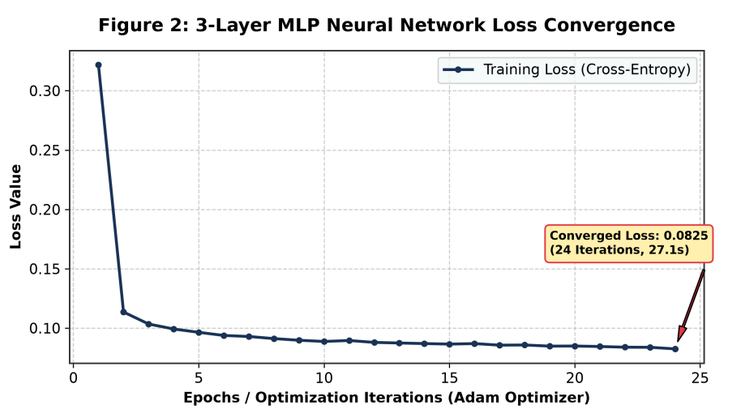


Figure 4. Training loss convergence curve of the real-time neural network over 24 epochs using Adam optimization.

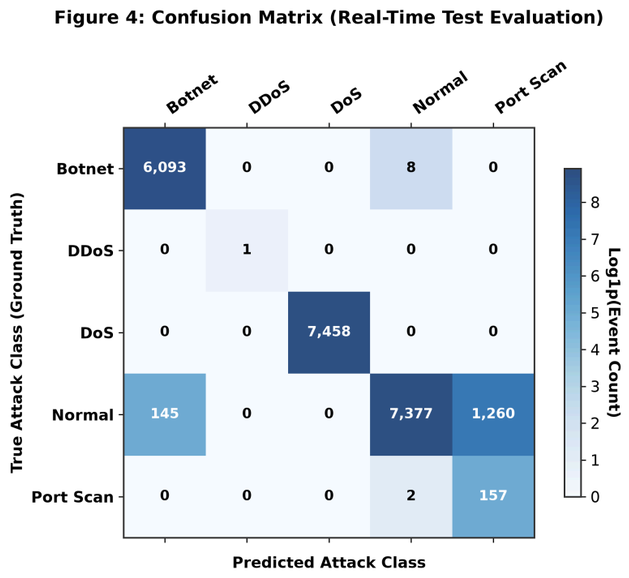


Figure 5. Confusion matrix across 22,501 test instances showing robust multi-class separation of DoS, Botnets, and Port Scans.

SOAR & Live Verification

- **4-Tier Cascading LLM Gateway:** NVIDIA NIM (Llama-3.2) → OpenAI GPT-4o → Google Gemini → Local Deterministic Rule Fallback.
- **Auto Firewall Containment:** Dynamically generates OS-specific mitigation scripts (Linux `iptables DROP`, Windows `netsh advfirewall`).



Figure 6. React 18 Glassmorphic SOC Console: Live threat radar, continuous risk scoring, and MITRE ATT&CK kill-chain mapper.

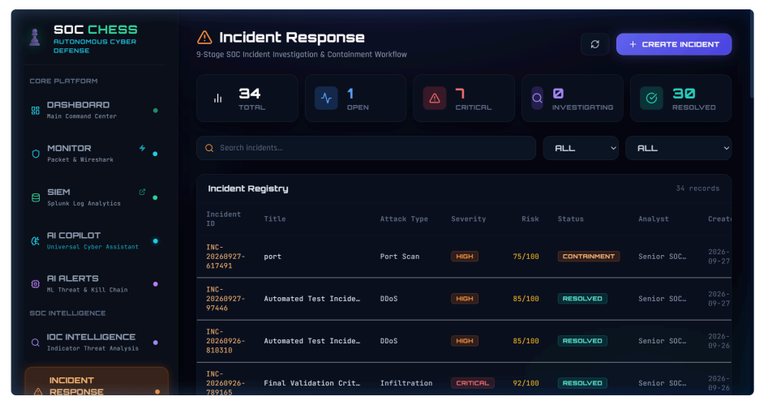


Figure 7. 5-Stage SOAR Incident Response Playbook executing dynamic firewall script isolation on active threat IP.

Conclusions

- **Unified Telemetry:** Successfully bridged live PCAP dissection and enterprise SIEM indexes into a synchronized command center.
- **Exceptional Precision:** Champion Random Forest achieved **99.39% accuracy** and MLP achieved **< 1.2 ms** latency, effectively solving alert fatigue.
- **Future Work:** Integrate kernel-level **eBPF / DPDK** for 10Gbps line-rate packet capture and Graph Neural Networks (GNNs) for Active Directory attack graphs.

References

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